- Free particle
- Freely falling particle under gravity
- Simple pendulum
- Double pendulum
- Pendulum where the pivot point moves in horizontal axis
- Atwood's machine

The Hamilton's principle can not be derived within classical mechanics. It can be motivated from the symmetry between space and time, but one can not formally prove it. One can find justification to the principle from quantum mechanics: In quantum mechanics each path in a transition contribute an amplitude $A \sim e^{iS/\hbar}$. Therefore, all but paths for which $\delta S = 0$ interfere destructively. One can view the Hamilton's principle as the $\hbar \to 0$ limit of the path integral formalism in quantum mechanics.

3.2 Variational Calculas

The Hamilton's principle (24) is equivalent to the mathematical problem in *calculus of variation.* In one dimension, the problem of calculus of variation is as follows: suppose there is a path $y = y(x)$ between two fixed end points x_1 and x_2. There is a functional $f(y, y', x)$, where $y' = dy/dx$, as a function of the path. In the calculus of variation, one finds the path $y = y(x)$ such that the integral

$$J = \int_{x_1}^{x_2} f(y(x), y'(x), x)dx\,, \tag{27}$$

is stationary. Here x acts as the parameter of the path. The meaning of stationary is the following: suppose we consider infinitesimally small variation in the neighbourhood the original path, then $\delta J = 0$ to first order in the varoation. We can denote a neighbourhood path as $y(x) + \alpha\eta(x)$ such that α is a infinitesimally small parameter and $\eta(x_1) = 0, \eta(x_2) = 0$. To be specific all paths can be written as

$$y(x, \alpha) = y(x, 0) + \alpha\eta(x)\,, \tag{28}$$

such that $\alpha = 0$ correspond to the original path. The required properties of the function $\eta(x)$ is that it is continuous and nonsingular and has continuous first and second derivative between x_1 and x_2.

Our aim is to find the actual path for which the value of the integral (27) is extremum – *i.e.*, for an infinitesimal deviation of the path from the original path, the deviation δJ. Since an infinitesimally close path to the original path is parametrized by a non-zero value of α one can write Hamilton's principle as

$$\left.\frac{dJ}{d\alpha}\right|_{\alpha=0} = 0\,.$$

Differentiating (22) w.r.to α we get (from now onwards we omit $|_{\alpha=0}$ sign for simplicity)

$$0 = \left.\frac{dJ}{d\alpha}\right|_{\alpha=0} = \int_{x_1}^{x_2} \frac{\partial f}{\partial \alpha}dx = \int_{x_1}^{x_2}\left(\frac{\partial f}{\partial y}\frac{\partial y}{\partial \alpha} + \frac{\partial f}{\partial y'}\frac{\partial y'}{\partial \alpha} + \frac{\partial f}{\partial x}\frac{\partial x}{\partial \alpha}\right)dx\,.$$

We have used Leibniz formula of differentiating an integration. Though we have used partial derivative inside the integration, we remember that since α is independent of x we could have used $d/d\alpha$ as well. Since x and α are independent, the last term within the bracket vanishes. The second term can be written as

$$\int_{x_1}^{x_2} \frac{\partial f}{\partial y'}\frac{\partial y'}{\partial \alpha}dx = \int_{x_1}^{x_2} \frac{\partial f}{\partial y'}\frac{d}{dx}\left(\frac{\partial y}{\partial \alpha}\right)dx\,,$$

where we have interchanged d/dx and $\partial/\partial\alpha$ since α and x are independent. Further integrating by parts we obtain

$$\int_{x_1}^{x_2} \frac{\partial f}{\partial y'}\frac{d}{dx}\left(\frac{\partial y}{\partial \alpha}\right)dx = \left.\frac{\partial f}{\partial y'}\frac{\partial y}{\partial \alpha}\right|_{x_1}^{x_2} - \int_{x_1}^{x_2} \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right)\frac{\partial y}{\partial \alpha}dx\,.$$

Since all the paths have the common end-points $(x_1, y(x_1))$ and $(x_2, y(x_2))$, and the end points are always fixed, the first terms is zero in the above line check yourself. Hence we get

$$0 = \left.\frac{dJ}{d\alpha}\right|_{\alpha=0} = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y} - \frac{\partial}{\partial x}\frac{\partial f}{\partial y'}\right)\frac{dy}{d\alpha}dx = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y'}\right)\eta(x)dx\,.$$

Since $y \equiv y(x, \alpha)$ are all independent paths, their variations $\eta(x) = dy/d\alpha$ are independent of each other. From the "fundamental lemma" of calculus it implies that the terms in the right-hand-side of the parenthesis vanishes

$$\boxed{\frac{\partial f}{\partial y} - \frac{d}{df}\left(\frac{\partial f}{\partial y'}\right) = 0}\,. \tag{29}$$

This is known the Euler-Lagrange equation. Solution of the equation yields the path $y = y(x)$ for which (27) is stationary. In other words, the solution of the EL equation (29) gives the path $y = y(x)$.

3.2.1 The brachistochrone problem

The statement of the problem is as follows: there is a particle that falls under the influence of gravity from a higher point (shown in figure as 1) from rest to a lower point (point 2). The problem is the find the path for which the time taken is minimum.

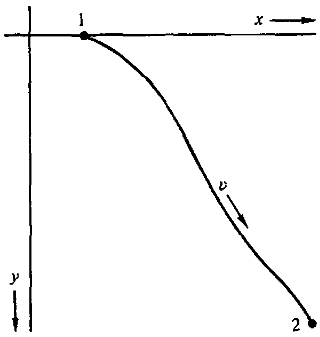

Figure 3: The brachistochrone problem.

If $v(t)$ is the instantaneous velocity of the particle and it travels a path ds in the interval dt then the total time taken is

$$t_{12} = \int_1^2 \frac{ds}{v} = \int_1^2 \frac{\sqrt{dx^2 + dy^2}}{v}\,. \tag{30}$$

The origin of the coordinate is at the point 1 and the y-axis vertically downwards. According to the conservation theorem

$$\frac{1}{2}mv^2 = mgy\,,$$

from which the velocity can be obtained as $v = \sqrt{2gy}$. Hence, the integral t_{12} can be written as

$$t_{12} = \int_1^2 \frac{\sqrt{1 + y'^2}}{\sqrt{2gy}}dx\,.$$

The integral t_{12} can be identified with J and the function f is

$$f = \sqrt{\frac{1 + y'^2}{2gy}}\,,$$

so that it satisfies the equation (26)

$$
\begin{aligned}
&-\frac{\sqrt{1+y'^2}}{2y\sqrt{y}}-\frac{d}{dx}\left(\frac{1}{\sqrt{y}}\cdot y'\cdot\frac{1}{\sqrt{1+y'^2}}\right)=0\,,\\
&\implies -2yy''=1+y'^2\,,
\end{aligned}
$$

where y'' double derivative of y with respect to x. After multiplying both sides by y' and then rearranging, it can be integrated as

$$
\begin{aligned}
\int\frac{2y'y''}{1+y'^2}=-\int\frac{y'}{y}\,,\quad &\implies \ln(1+y'^2)=-\ln y+A\,,\\
&\implies 1+y'^2=\frac{B}{y}\,,
\end{aligned}
$$

where $B=e^A$. We need to integrate it one more time. To this end we write the last line above as

$$
\frac{\sqrt{y}dy}{\sqrt{B-y}}=\pm dx\,.
$$

To integrate it we make a change in variable as $y=B\sin^2\phi$ so that $dy=2B\sin\phi\cos\phi d\phi$. Integrating we get

$$
B(2\phi-\sin 2\phi)=\pm 2x-C\,,
$$

where C is a constant. Redefining $\theta=2\phi$ we get the parametric solutions as

$$
x=\pm a(\theta-\sin\theta)\pm d\,,\quad y=a(1-\cos\theta)\,,
$$

where $a=B/2$ and $d=C/2$. Using the initial condition $(x,y)=(0,0)$ yields $d=0$. On the other hand we assume that x always increase during the motion, so that only positive sign can be chosen. Hence the final solution is

$$
x=a(\theta-\sin\theta)\,,\quad y=a(1-\cos\theta)\,. \tag{31}
$$

This is the parametric equation of a curve that is called a cycloid.
Practice problems: shortest distance between two points in a plane is a straight line, shape of a wire hanging freely under its own weight,

Some other problems that will be discussed in class or in tutorials

- Suppose there is a curve in the $x-y$ lane between two fixed points (x_1,y_1) and (x_2,y_2). The plane is revolved around the y-axis. The problem is to find the curve for which the surface area is minimum.
- The catenary curve
- Geodesic on a given geometry

3.3 Conservation theorems

3.3.1 Conjugate/Canonical/generalized momentum

Consider a system in motion under a force that can be derived from a potential, which is a function of position alone. To describe the system, we use Cartesian coordinates $x_i=(x,y,z)$. For example, this could represent the motion of a particle under the influence of gravity. In this context, the first term in equation (26) corresponds to

$$
\frac{\partial L}{\partial x_i}=\frac{\partial T}{\partial x_i}-\frac{\partial V(x,y,z)}{\partial x_i}=-\frac{\partial V(x,y,z)}{\partial x_i}=F_{x_i}\,,
$$

since T is independent of x_i. From the second term we get

$$\begin{aligned}\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}_i}\right) &= \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{x}} - \frac{\partial V(x,y,z)}{\partial \dot{x}}\right), \\ &= \frac{d}{dt}\left(\frac{\partial}{\partial \dot{x}}\sum \frac{m^2}{2}(\dot{x}_i^2)\right) = \frac{d}{dt}(m_i\dot{x}_i) = \frac{d}{dt}(p_i)\,.\end{aligned}$$

This leads to

$$\frac{d}{dt}(p_i) = F_{x_i}\,.$$

If F_{x_i} is interpreted as the force, p_i the linear momentum and the above equation is Newton's 2nd law of motion. The interesting part of the result is that the linear momentum is derived from the Lagrangian as $p_i = \frac{\partial L}{\partial \dot{q}_i}$. The result can be further generalized to define *generalized momentum* or the *canonical momentum* or *conjugate momentum* for the coordinate q_i as

$$p_i = \frac{\partial L}{\partial \dot{q}_i}\,. \tag{32}$$

If the coordinates q_i are not Cartesian, the corresponding momenta p_i may not have the dimensions of linear momentum. Furthermore, for velocity-dependent potentials, even in the case of Cartesian coordinates, the generalized coordinates q_i are not necessarily identical to the mechanical momentum. An example of this is the motion of a charged particle in an uniform electromagnetic field, where the Lagrangian is

$$L = \frac{m^2}{2}\dot{r}^2 - Q\phi + Q\vec{A}.\dot{\vec{r}},$$

where $\vec{r}$ is the position vector of the particle, Q is the charge, and $\vec{A}$ is the magnetic vector potential. In this example the x-component of the mechanical momentum of the particle is $m\dot{x}$, but the generalized momentum is

$$p_x = \frac{\partial L}{\partial \dot{x}} = m\dot{x} + QA_x\,.$$

If the Lagrangian does not depend on a given coordinate q_i, but depends on its corresponding velocity $\dot{q}_i$, then the coordinate is said to be *cyclic* (or *ignorable*). In this case, the Euler-Lagrange equation implies that $\frac{\partial L}{\partial q_i} = 0$, and the equation of motion becomes

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = 0 \Longrightarrow p_i = \text{constant}\,,$$

so, *generalized momentum corresponding to the generalized coordinate is conserved.* In the previous example, of both ϕ and $\vec{A}$ are independent of r then

$$p_x = \frac{\partial L}{\partial \dot{x}} = m\dot{x} + QA_x = \text{constant}\,.$$

Here, what is conserved is not the mechanical linear momentum $m\dot{x}$ but sum of $m\dot{x}$ and QA_x.

3.3.2 Energy conservation

Consider a Lagrangian that has no explicit time dependence, *i.e.,*

$$\frac{\partial L}{\partial t} = 0\,.$$

Its total time derivative is

$$\begin{aligned}\frac{dL}{dt} &= \sum_i \frac{\partial L}{\partial q_i}\dot{q}_i + \sum_i \frac{\partial L}{\partial \dot{q}_i}\frac{d\dot{q}_i}{dt}\,, \\ &= \sum_i \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right)\dot{q}_i + \sum_i \frac{\partial L}{\partial \dot{q}_i}\frac{d\dot{q}_i}{dt}\,, \\ &= \sum_i \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\dot{q}_i\right) = \sum_i \frac{d}{dt}\Big(p_i\dot{q}_i\Big)\,.\end{aligned}$$

Combining the two sides of the above equation we get

$$\frac{d}{dt}\left(\sum_i p_i\dot{q}_i - L\right) = 0\,. \tag{33}$$

Since the total time derivative vanishes for the function $\sum_i p_i\dot{q}_i - L$, it is is a constant of motion. It is called the energy function

$$h(q_i, \dot{q}_i; t) = \sum_i p_i\dot{q}_i - L\,. \tag{34}$$

It can be shown that if L has no explicit time dependence, and the potential is a function of the position coordinates alone, then the energy function is equal to the total energy of the system.

Examples: Simple pendulum, double pendulum, coupled pendulum, etc

3.4 Non-uniqueness of Lagrangian

Any given Lagrangian is unique *up to the addition of a total time derivative* of a function of the coordinates and time. That is, if $L(q, \dot{q}, t)$ is a Lagrangian and one defines a new Lagrangian by adding the total time derivative of a function $F(q, t)$,

$$L' = L + \frac{dF}{dt}, \tag{35}$$

then L' yields the same equations of motion as the original Lagrangian L. This follows from the relation between the corresponding actions. The new action is

$$\begin{aligned} S' &= \int_{t_1}^{t_2} L'\,dt \\ &= \int_{t_1}^{t_2}\left(L + \frac{dF}{dt}\right)dt \\ &= S + F(q_2, t_2) - F(q_1, t_1), \end{aligned} \tag{36}$$

where $q_1 = q(t_1)$ and $q_2 = q(t_2)$. Since the endpoints of the variation are fixed, $\delta q(t_1) = \delta q(t_2) = 0$, the boundary terms do not contribute to the variation of the action. Therefore,

$$\delta S' = \delta S, \tag{37}$$

and the Euler–Lagrange equations derived from L' are identical to those derived from L.

The same result can also be verified directly by substituting L' into the Euler–Lagrange equations and observing that the additional terms cancel identically.

3.5 Symmetry and Noether's theorem

Noether's Theorem (Classical Mechanics).

Consider a dynamical system with generalized coordinates $q_i(t)$ and Lagrangian $L(q_i, \dot{q}_i, t)$. Suppose the action

$$S = \int_{t_1}^{t_2} L(q_i, \dot{q}_i, t)\,dt$$

is invariant under a continuous one-parameter transformation

$$q_i(t) \to q_i'(t') = q_i(t) + \epsilon\,\delta q_i(q, t),$$

up to a total time derivative,

$$L'(q', \dot{q}', t') = L(q, \dot{q}, t) + \epsilon\,\frac{dF}{dt},$$

where ϵ is an infinitesimal parameter and $F(q,t)$ is some function of coordinates and time. Then according to Noether's theorm there exists a conserved quantity, called Noether charge

$$Q = \sum_i \frac{\partial L}{\partial \dot{q}_i}\,\delta q_i - F,$$

which satisfies

$$\frac{dQ}{dt} = 0$$

along solutions of the Euler–Lagrange equations.

Proof: The proof is shown only for one coordinate transformation – generalization to a set is left as an exercise. If the action

$$S = \int_{t_1}^{t_2} L(q,\dot{q},t)dt\,,$$

is invariant under coordinate transformnation

$$q(t) = q(t) + \alpha\delta q\,,$$

where α is an infinitesimally small parameter. The change in velocity is

$$\delta\dot{q} = \delta\left(\frac{dq}{dt}\right) = \frac{d}{dt}\delta q\,, \tag{38}$$

The last line follows from the fact that the coordinate transformation is not time dependent. The resultant change in the Lagrangian is

$$\begin{aligned}
\delta L &= \frac{\partial L}{\partial q}\alpha\delta q + \frac{\partial L}{\partial \dot{q}}\alpha\delta\dot{q}\,, \\
&= \frac{\partial L}{\partial q}\alpha\delta q - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right)\alpha\delta q + \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\alpha\delta q\right), \\
&= \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\alpha\delta q\right).
\end{aligned} \tag{39}$$

There are two possibilities for which the action S can remain invariant, either $\delta L = 0$ or $\delta L = \alpha\frac{dF}{dt}$. So in general

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\alpha\delta q - \alpha F\right) = 0$$

which implies that

$$Q = \frac{\partial L}{\partial \dot{q}}\delta q - F \tag{40}$$

is conserved. The Q is known as Noether charge. If there are a set of coordinates $\{q_i\}$ then the conserved charge is

$$Q = \sum_i \frac{\partial L}{\partial \dot{q}_i}\delta q_i - F\,. \tag{41}$$

Example 1: Free particle and the conservation of linear momentum. The conservation follows from the fact that the free particle Lagrangian

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)\,,$$

the action is symmetric under translation

$$x \to x + \epsilon, \quad \delta x = 1, \delta t = 0\,.$$

The change in the Lagrangian is $\delta L = 0$. So the conserved charge is

$$Q = \frac{\partial L}{\partial \dot{x}} 1 = m\dot{x}\,.$$

So

$$\boxed{\text{translation symmetry leads to conservation of linear momentum}}$$

Example 2: Rotational symmetry and conservation of linear momentum. exercise